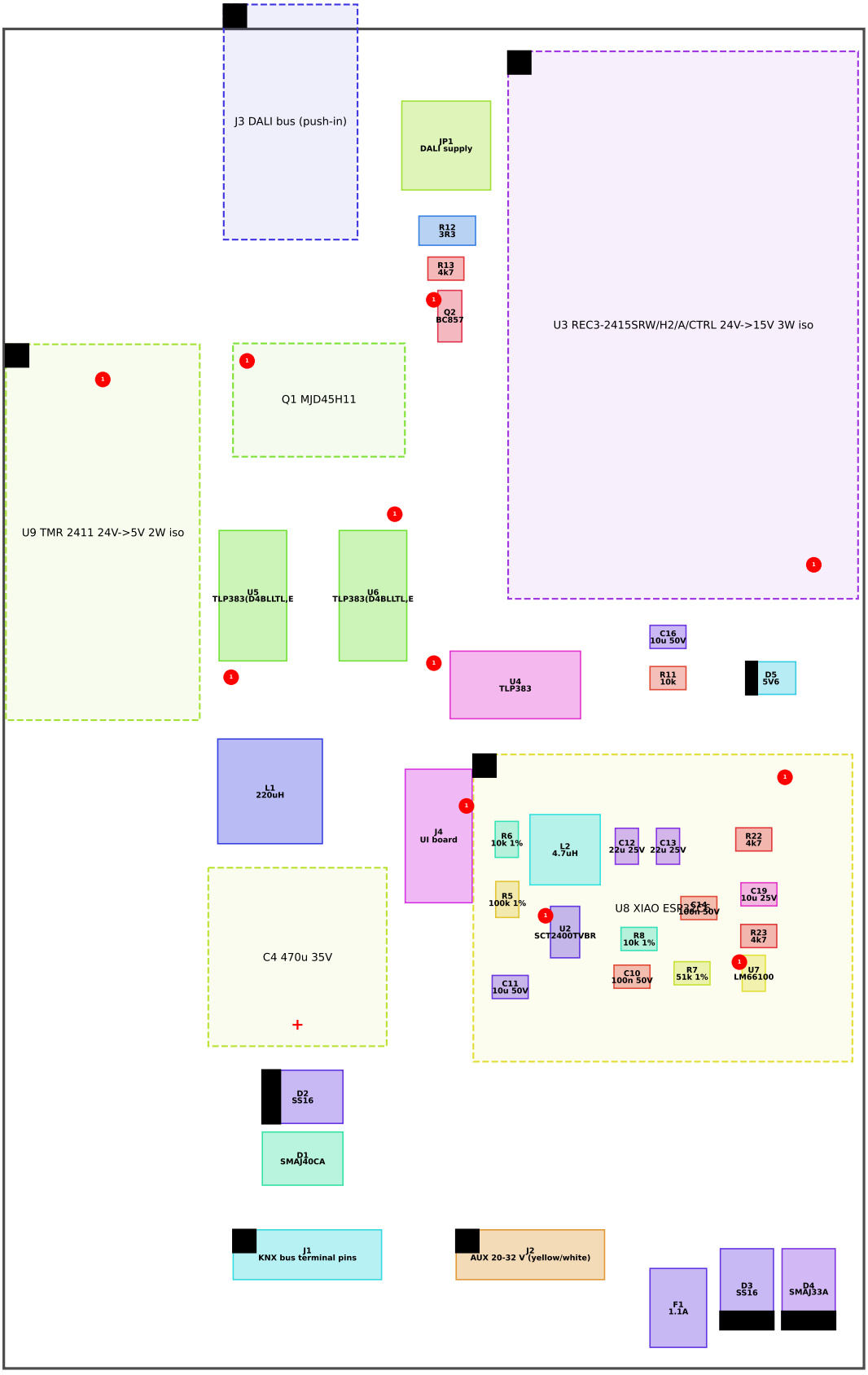
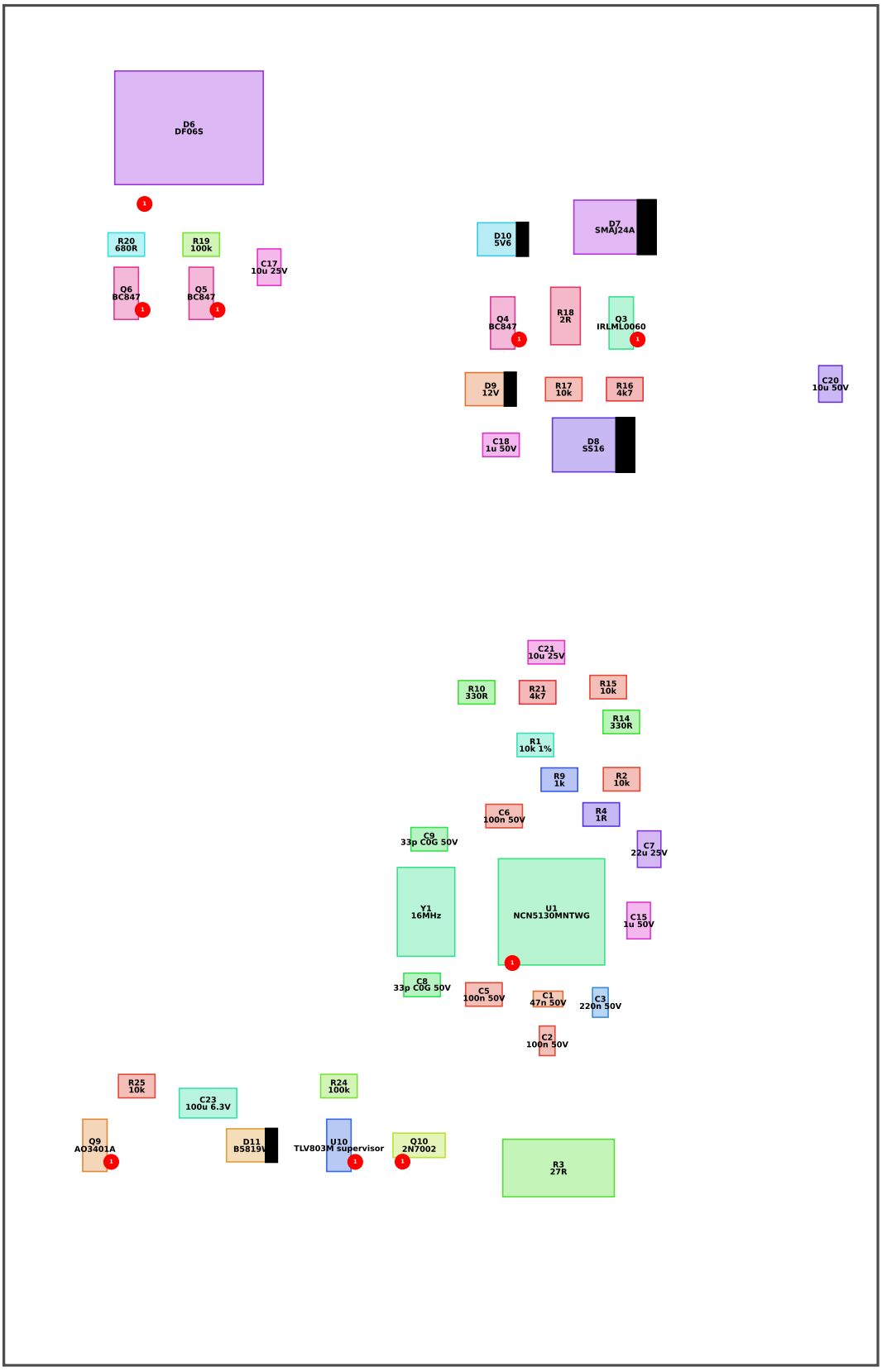
main - top side

black bar = diode cathode, "+" = electrolytic +, red "1" = pin 1, no mark = either way;
black corner = hand-fitted, dashed = large part, colour = value (key on the last pages)



main - bottom side (mirrored, seen from below)
black bar = diode cathode, "+" = electrolytic +, red "1" = pin 1, no mark = either way;
black corner = hand-fitted, dashed = large part, colour = value (key on the last pages)



main - values (85 parts)

100k	x2	bottom	R19, R24
100n 50V	x3	bottom	C2, C5, C6
100u 6.3V	x1	bottom	C23
10k	x4	bottom	R2, R15, R17, R25
10k 1%	x1	bottom	R1
10u 25V	x2	bottom	C17, C21
10u 50V	x1	bottom	C20
12V	x1	bottom	D9
16MHz	x1	bottom	Y1
1R	x1	bottom	R4
1k	x1	bottom	R9
1u 50V	x2	bottom	C15, C18
220n 50V	x1	bottom	C3
22u 25V	x1	bottom	C7
27R	x1	bottom	R3
2N7002	x1	bottom	Q10
2R	x1	bottom	R18
330R	x2	bottom	R10, R14
33p C0G 50V	x2	bottom	C8, C9
47n 50V	x1	bottom	C1
4k7	x2	bottom	R16, R21
5V6	x1	bottom	D10
680R	x1	bottom	R20
AO3401A	x1	bottom	Q9
B5819W	x1	bottom	D11
BC847	x3	bottom	Q4, Q5, Q6
DF06S	x1	bottom	D6
IRLML0060	x1	bottom	Q3
NCN5130MNTWG	x1	bottom	U1
SMAJ24A	x1	bottom	D7
SS16	x1	bottom	D8
TLV803M supervisor	x1	bottom	U10
1.1A	x1	top	F1
100k 1%	x1	top	R5
100n 50V	x2	top	C10, C14
10k	x1	top	R11
10k 1%	x2	top	R6, R8
10u 25V	x1	top	C19
10u 50V	x2	top	C11, C16
220uH	x1	top	L1

	22u 25V	x2	top	C12, C13
	3R3	x1	top	R12
	4.7uH	x1	top	L2
	470u 35V	x1	top	C4
	4k7	x3	top	R13, R22, R23
	51k 1%	x1	top	R7
	5V6	x1	top	D5
	AUX 20-32 V (yellow/white)	x1	top	J2
	BC857	x1	top	Q2
	DALI bus (push-in)	x1	top	J3
	DALI supply	x1	top	JP1
	KNX bus terminal pins	x1	top	J1
	LM66100	x1	top	U7
	MJD45H11	x1	top	Q1
	REC3-2415SRW/H2/A/CTRL 24V->15V 3W iso	x1	top	U3
	SCT2400TVBR	x1	top	U2
	SMAJ33A	x1	top	D4
	SMAJ40CA	x1	top	D1
	SS16	x2	top	D2, D3
	TLP383	x1	top	U4
	TLP383(D4BLLTL,E	x2	top	U5, U6
	TMR 2411 24V->5V 2W iso	x1	top	U9
	UI board	x1	top	J4
	XIAO ESP32C6	x1	top	U8

main - how to read this drawing and the parts

Black bar = diode cathode (K). The diode body carries a band on that end.

"+" = electrolytic capacitor + pad. The can has a dark stripe (or "-" marks) on the other side.

Red dot "1" = pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules:

- SOT-23 / SOT-23-5 / TSOT-23-6 / SC-70-6: pin 1 is marked by a dot or bevel, or follow the leg pattern (SOT-23: the single leg is pin 3, opposite pins 1 and 2).
- SO-6 optocouplers, the NCN5130 (QFN) and the supervisor: a dot or notch at pin 1 on the package.
- WS2812B-MINI-V3J: a notch / marked corner on the body is pin 1 (VDD); the legs differ, so the footprint only fits one way.
- The XIAO, DC-DC converters and the REC3 module: their silkscreen / pin-1 marking and the socket shape.

No mark = fits either way (resistors, ceramic capacitors, inductors, fuse, crystal, pin strips).

Black corner = fitted by hand (not placed by the assembler); dashed outline = large part.

Same colour = same value (see the values page).

Tools: a 3-5x head magnifier or a 10x loupe and good light are enough to read the bands and pin-1 dots on SOT-23 and 0805 parts; a USB microscope helps for the NCN5130 QFN and SC-70-6.